Automatic Category Suggestion for Wikimedia Commons

Milestone 2 — Progress Report | Team UG1

1. Approach and changes since M1

We treat category suggestion as **open-vocabulary extreme multi-label retrieval**, as the problem statement frames it. A pretrained vision–language encoder maps each image to an L2-normalized embedding, and categories are treated as retrievable objects rather than a fixed softmax. Four complementary scorers produce per-category scores: (i) **kNN retrieval** over a FAISS index of labelled images, (ii) **zero-shot text prototypes** from the encoder's text tower, (iii) a **supervised multi-label head** (asymmetric loss, text-initialized), and (iv) a **metadata** scorer over title/description. A calibrated logistic **fusion** combines the four, and a category-hierarchy **DAG** suppresses redundant ancestors (most-specific rule). The M1 design is intact; work since M1 has strengthened every stage and fixed defects that were suppressing accuracy:

- **Encoder upgraded** from CLIP ViT-B/32 to a **pluggable backbone**; the default is **SigLIP-so400m** (state-of-the-art zero-shot) on GPU, with CLIP ViT-L/14 and OpenCLIP ViT-H/14 selectable in one line. Embedding dim is inferred and caches are namespaced per backbone.
- **Prompt ensembling** (14 templates averaged) for the text prototypes — a known zero-shot boost that also improves the head's text initialization and the metadata scorer.
- **Data cleaning per spec:** hidden/maintenance & licence categories removed at source (`clshow=!hidden`); temporal train/test split; exact + embedding-cosine cross-split **near-duplicate removal** to prevent leakage.
- **Collection hardened:** a single retrying HTTP **session with** `maxlag` replaces per-call requests (fixing severe API throttling), and **breadth-first subcategory traversal** fills container categories (e.g. *Birds, Fishes*) that have no direct files.
- **Fusion rebuilt:** leave-one-out kNN features (removes train self-retrieval leakage) + feature standardization (fixes degenerate near-constant outputs).
- **Supervised head** now checkpoints on validation **ranking (hit@3)** rather than loss, preventing rank collapse on imbalanced data.
- **Evaluation extended:** hierarchy-aware precision, robustness-by-degradation bands, and latency p50/p95, in addition to P@k / R@k / F1@k / mAP with head–torso–tail tiers.

2. Experiments and results

All metrics use a **temporal hold-out** (train earlier, test later) with maintenance categories removed. Table 1 shows per-scorer precision on a 56-category hold-out (CLIP ViT-L/14); the fused row is from the corrected system on an 8-category SigLIP controlled hold-out, whose full metrics are in Table 2.

Scorer	P@1	P@3	P@5
kNN retrieval	0.15	0.11	0.10
Zero-shot text (ensembled)	0.79	0.32	0.20
Supervised head	0.05	0.30	0.19
Metadata	0.37	0.17	0.11
Fusion — *before* fix	0.14	0.10	0.09
Fusion — *after* fix	**0.97**	**0.50**	**0.30**

Table 1. Per-scorer P@k on a 56-category temporal hold-out (CLIP ViT-L/14). The fused "after fix" row is measured on the 8-category SigLIP controlled hold-out of Table 2.

Accuracy (top-1)	mAP	R@1	R@3	F1@1	F1@3	F1@5
0.97	0.94	0.70	0.97	0.79	0.64	0.45

Table 2. Full fused-system metrics on the 8-category SigLIP controlled temporal hold-out (accuracy = top-1). R@3 = 0.97, R@5 = 0.97, mAP = 0.94.

Key findings.

- The **zero-shot text scorer alone reaches P@1 ≈ 0.79**, confirming the encoder captures image content well. The training-dependent scorers (kNN, supervised) scored low only because an **image-download bug fetched ~5% of images** (120 of 2,872); with the robust session this is ~100% in validation, which directly lifts those scorers.
- The **fusion was the main defect**: it collapsed to a near-constant ~0.10 (the positive base rate) and even gave the *text* scorer a **negative** weight, discarding the best signal — the "confident but unrelated" behaviour we observed. Causes: (a) fitting on the same set the kNN index holds, so every train query retrieved itself; (b) an under-regularized, unstandardized logistic fit. **After the fix**, weights are all positive (e.g. [kNN 3.0, head 0.74, text 0.72, meta 0.22]), scores span 0.001–0.998, and the fused system matches or beats the strongest single scorer (Table 2).
- **Collection throttling diagnosed:** direct API experiments showed plain per-call requests are throttled after ~5 rapid calls (empty responses), which had collapsed one run's usable vocabulary to **4 classes**; a persistent session with `maxlag` returns 100% success. Container categories (Birds, Fishes) returned 0 direct files — now handled by subcategory BFS.

- **Latency harness** reports ~68 ms/image (p50) on CPU with the small backbone; full-scale GPU numbers and FAISS memory estimates are produced automatically by the notebook.

3. Progress on testing

- **End-to-end execution** of the whole notebook via `nbconvert` after every change; the current pipeline runs with **0 cell errors** on both the CLIP and SigLIP backends.
- **Targeted validation per fix:** fusion leave-one-out (weight/score sanity), backbone-switch cache test (dimension-safe, no re-download), collection throttling (HTTP status under load), and SigLIP integration (text-padding and embedding path). A **Flask demo UI** was verified end-to-end (Aircraft 0.99, Beaches 0.99, Cats 0.84, Dogs 0.94).
- **Metrics implemented and exercised:** P@k, R@k, F1@k, macro mAP, head/torso/tail/zero-shot tiers, hierarchy-aware precision, robustness bands, and latency p50/p95.

4. Plan for the remaining time

- **Full-scale run** (now unblocked): ~100 seed categories × 200 images via subcategory BFS → target several hundred classes and ~15k images; report full temporal-holdout metrics with tier stratification and a **reviewed-score** (human-checked false-negative) estimate.
- **Hierarchical taxonomy & classes** — our biggest planned lever: (i) build the **full multi-level Commons category DAG** (beyond immediate parents) by crawling `categorylinks`; (ii) **hierarchical label expansion** during training — propagate each file's labels to its ancestors to counter the systematic unlabelled-ancestor bias; (iii) **taxonomy-constrained inference** enforcing ancestor consistency and the most-specific rule; (iv) **coarse-to-fine routing** (predict a top-level family, then retrieve within it) to scale toward millions of categories; and (v) **hierarchical evaluation** — lowest-common-ancestor (LCA) precision/recall and per-depth metrics, so near-misses on the tree are scored fairly.
- **Backbone benchmark at scale:** SigLIP-so400m vs CLIP ViT-L/14 vs OpenCLIP ViT-H/14.
- **Continual learning:** feedback loop (accept/reject → hard negatives) with scheduled head retraining; uploader-grouped splits to prevent style leakage.
- **Serving & scale:** PQ/HNSW FAISS index and INT8/ONNX encoder to validate the 10^8-scale memory and latency estimates.
- **Accuracy:** multilingual category aliases in the prompt ensemble, and per-tier fusion thresholds for better tail behaviour.

Code: the entire pipeline is implemented in a single reproducible notebook (`train.ipynb`), with a minimal Flask upload UI (`webapp/`); runnable on Kaggle (P100) or locally.